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Monetary growth and wealth inequality[☆]

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ABSTRACT

We construct a global panel data set to examine the relationship between monetary growth and wealth inequality. Dynamic panel estimates suggest that both overall and inherited wealth inequality increase with growth in the base money supply. These results hold for countries with at least one billionaire and for OECD countries.

1. Introduction

Wealth inequality is returning to levels not seen since World War I, with the top decile in the US controlling over 70 percent of wealth (Piketty, 2014). Some argue that monetary inflation spurs wealth inequality, yet less attention is devoted to uncovering the empirics behind such a theory (Hülsmann, 2014). Often income inequality is chosen as a substitute for wealth inequality due to data limitations (Balac, 2008). Only a handful of studies consider the link between wealth inequality and monetary policy (Doepke and Schneider, 2006; Doepke et al., 2015; Meh et al., 2010; Schroth, 2016; Curran and Dressler, 2019). They each model interest rate channels in the US.

Ours is the first empirical examination of the relationship between monetary growth and wealth inequality. The dataset of Bagchi and Svejnar (2015) provides wide country coverage and allows a decomposition of various types of wealth inequality. We find that expansionary monetary policy explains in part the rise in wealth inequality. We explain this relationship through Cantillon effects and test for empirical evidence of this connection using dynamic panel data estimation. Modern conventional monetary policy has adverse distributional effects on wealth in line with Cantillon, ([1755] 2015).

2. Theory and method

Let us adapt the argument of Cantillon, ([1755] 2015) to the environment of fiat currency. When open market operations increase the money supply, that new money ends up in the reserves of banks, who then have the opportunity to lend this new money. While being the first to receive these loans of new money does not increase one's wealth, it allows one to bid up the price of capital goods (Hülsmann, 2014). In this way, the wealthy, having better access to credit (Robb and Fairlie, 2007), are able to increase their purchasing power. Banks increase wealth inequality by funneling new money into the investments of the wealthy, whereas the poor receive this new money after its purchasing power has been eroded.

Our dependent variable is wealth inequality measured by billionaire wealth as a percentage of GDP (B/GDP), which also enters as an explanatory variable with a lag to control for long lagged effects between the independent variables and the dependent variable. Because the key argument is that those receiving new money first benefit from the expansion of the money supply, the monetary variables are not lagged.[1] Base money growth (BMG) is chosen for three reasons. First, base money growth is more under control of the central bank than other measures and thus better captures the essence of monetary policy than monetary variables that are the result of bank credit creation. While such variables may also have Cantillon effect relationships, they test the relationship between a fractional reserve banking system and

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1 *Price* inflation does not capture relative price changes and only measures overall changes after *monetary* inflation has occurred.

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inequality rather than the relationship between monetary policy and inequality. Second, base money growth is mentioned directly by Hülsmann (2014) as a cause of redistribution. Third, the IMF's coverage of base money growth rates is wide.

Control variables are education as measured by the UN education index (EDU), taxes as a percentage of GDP (TAX), and logged GDP per capita (LNGDP). Education increases social mobility, which is negatively correlated with inequality (OECD, 2010). Measured with a two year lag, education likely has a lagged interaction with inequality as it takes time to build human capital. Higher taxation is expected to reduce inequality as taxes are used to finance social expenditures and lower the incomes, and hence over time, the wealth of those at the top of the wealth distribution. As an alternative measure of monetary policy, the central bank target rate is substituted for base money growth. The empirical specification is

$$\left(\frac{B}{GDP}\right)_{it} = \beta_0 + \beta_1 BMG_{it} + \beta_2 \left(\frac{B}{GDP}\right)_{it-1} + \beta_3 EDU_{it} + \beta_4 TAX_{it} + \beta_5 LNGDP_{it} + \varepsilon_{it} \quad (1)$$

where i indexes country and t indexes annual time period.[2]

We estimate a dynamic model as current wealth inequality likely depends on its own lag. Monetary policy has considerable lags with its impact on both prices and output (Romer and Romer, 2004). In addition, Cantillon effects argue that the effect is felt by those receiving new money first (Cantillon, [1755] 2015). We employ GMM to avoid Nickell bias. System GMM is the more efficient estimator because difference GMM loses too many observations in unbalanced datasets due to its method of differencing. Such analysis constrains the number of variables, because over-instrumentation is a concern, and we run tests on the regressions to make sure that the experiment is not over-instrumented (Roodman, 2009).

3. Data

Table S1 of the supplementary appendix presents summary statistics for all variables. Wealth inequality is measured using a proxy created by Bagchi and Svejnar (2015), billionaire wealth as a percentage of GDP. Data on billionaire wealth are compiled from *Forbes*' listing of billionaires. The authors classified the world's billionaires by origin of wealth in the years 1987, 1992, 1996, 2002, 2007, and 2012. We focus on two of the authors' classifications of wealth: overall and inherited. This dataset is the most comprehensive global wealth inequality dataset due to its decomposition into types of wealth inequality and wide coverage. For data sources on the explanatory variables, see the online appendix.

Our data set is an unbalanced panel covering 6 time periods and 143 countries. The subsample of countries with at least one billionaire covers 6 time periods and 57 countries. The OECD subsample covers 6 time periods and 34 countries. With the exception of the central bank targeted rate, no variables come from multiple sources. With the exception of the UN education index, all variables pertain to the year under observation.

[2] To adjust for cyclic variation in wealth, we included a dummy variable (REC) that takes a value of 1 if a country experienced negative GDP growth in one of the years between observations and 0 otherwise. This also controls for the countercyclicality of monetary policy. Results were robust and the recession dummy was always insignificant, thus we omit it in (1).

Table 1
Wealth inequality and base money growth.

Panel A *Wealth Inequality*	(1) One-step	(2) One-step with Robust SE	(3) Two-step with Robust SE
Base money growth	0.706***	0.706*	0.845***
	(2.96)	(1.84)	(3.86)
First lag of dependent variable	0.516***	0.516***	0.442**
	(5.37)	(3.03)	(2.06)
Education	0.068	0.068	0.112
	(0.37)	(0.36)	(0.76)
Tax as a percentage of GDP	0.256*	0.256	0.237
	(1.94)	(1.55)	(1.39)
Log GDP per capita	1.090	1.090	1.980
	(0.41)	(0.47)	(0.74)
Constant	−17.292	−17.292	−29.760
	(−0.78)	(−0.77)	(−1.29)
N	338	338	338
Sargan test statistic	0.046	–	–
Arellano–Bond I	–	0.013	0.107
Arellano–Bond II	–	0.690	0.711
Panel B *Inherited Wealth Inequality*	(1) One-step	(2) One-step with Robust SE	(3) Two-step with Robust SE
Base money growth	0.484***	0.484*	0.571***
	(4.95)	(1.90)	(3.31)
First lag of dependent variable	0.419***	0.419***	0.335*
	(5.88)	(2.62)	(1.78)
Education	−0.002	−0.002	0.008
	(−0.03)	(−0.03)	(0.11)
Tax as a percentage of GDP	0.043	0.043	0.003
	(0.86)	(1.06)	(0.08)
Log GDP per capita	−0.764	−0.764	−0.266
	(−0.90)	(−0.54)	(−0.33)
Constant	6.674	6.674	1.457
	(0.92)	(0.49)	(0.18)
N	337	337	337
Sargan test statistic	0.119	–	–
Arellano–Bond I	–	0.022	0.141
Arellano–Bond II	–	0.960	0.942

Notes: *Panel A*: dependent variable is billionaire wealth as a percentage of GDP. *Panel B*: dependent variable is *inherited* billionaire wealth as a percentage of GDP. Z statistics in parentheses. Education is the UN education index. Base money growth is from the IMF, while M1 is from the OECD. Tax revenue as a percentage of GDP is from the World Bank. GDP per capita is from the Penn World Tables. Arellano–Bond I and II are the Arellano–Bond tests for first-order and second-order autocorrelation.
*Significant at 10%.
**Significant at 5%.
***Significant at 1%.

4. Empirical results

4.1. Base money growth and wealth inequality

Table 1 reports the estimates of Eq. (1). Overall wealth inequality is the dependent variable in panel A, while inherited wealth inequality is the dependent variable in panel B. The first column reports one-step GMM estimates. Columns two and three report one- and two-step GMM estimates with robust standard errors.

Base money growth has a significant positive relationship with overall and inherited wealth inequality. The smaller coefficient for base money growth in panel B is to be expected because inherited billionaire wealth cannot exceed overall billionaire wealth. Lagged wealth inequality is significantly positive as expected. Other variables are insignificant.[3] We find evidence of Cantillon

[3] Taxes are only significant in column 1 of panel A. The Sargan test value suggests evidence, however, that the overidentifying restrictions may be invalid. With mild misspecification, robust standard errors in the two-step estimators correct for the downward bias of possible over-instrumentation (Roodman, 2009).

effects through the significant positive relationship between base money growth and wealth inequality. This relationship is most significant in the best specified model.[4]

4.2. Extensions

Section S2 of the online appendix examines the robustness of our results to countries with at least one billionaire, OECD countries, central bank target rates, income inequality, and the level of M1 money supply. Restricting attention to countries that have a billionaire in at least one of the years under observation, we find that our choice of sample did not drive our results for base money growth. Our results for base money growth in the OECD subsample confirm that our findings are robust to the level of economic development. We are wary of drawing any strong inference regarding the relationship between wealth inequality and central bank target rates because the significance of the results changes across columns. When including income inequality or the level of M1, the key variable becomes insignificant.

5. Conclusion

Employing data across a diverse sample of countries, ours is the first empirical analysis of the relationship between monetary growth and wealth inequality. There is a significant positive relationship between base money growth and wealth inequality, robust for both overall and inherited wealth inequality. Our findings are significant in a subsample of countries that have billionaires at least once and in OECD countries. These results should strengthen the case for making distributional effects a part of monetary policy conversation.

[4] The Wald test rejects the null hypothesis for no overall significance. First-order serial correlation is present, as is expected, but we find no evidence of second-order serial correlation.

Appendix A. Supplementary data

A supplementary appendix can be found at http://michael-curran.com/research/Inequality_SupplementaryAppendix_Long.pdf.

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Supplementary Appendix to: "Monetary Growth and Wealth Inequality"

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The structure of this appendix is as follows. Section S1 describes the data sources for the explanatory variables and displays a table of summary statistics for the variables employed in the analysis. Section S2 reports regression results on a number of robustness extensions.

S1 Data Sources and Summary Statistics

The key explanatory variable, base money growth, comes from the International Monetary Fund's website. The other control variables come from the UN, the World Bank, and the Penn World Tables (Feenstra, Inklaar and Timmer, 2015). Education is proxied by the UN education index, which is measured two years before the year under review. Data on tax revenue as a percentage of GDP come from the World Bank and is used as a measure of the amount of tax paid by citizenry. We use the Penn World Tables measures of GDP and population to construct a measure of GDP per capita.

The central bank target rate variable comes from calculations made using each country's central bank's website. The central bank target rate for all of the euro area countries is the European Central Bank (ECB) targeted rate. The data on central bank targeted rates are used only in the subsample of OECD countries. All euro area countries share the euro area monetary aggregates. Also factoring into the OECD regression is an alternate control variable, net Gini index, and the level of M1 money supply, both of which come from the OECD. The Gini coefficient measures income inequality ranging from 0 to 1, where 0 indicates that income is uniformly distributed in the population and 1 indicates the highest level of inequality possible.

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Table S1: Summary Statistics for Variables in Overall Sample, Subsample of Countries Which Have Billionaires at Least Once, and the OECD Subsample

Variables for Overall Sample	Mean	Median	Min	Max	Standard Deviation
Billionaire Wealth (% GDP)	0.02	1.85	0	217.85	8.80
Inherited Billionaire Wealth (% GDP)	0.01	0.70	0	16.44	2.01
Education	60.3	57.69	8.1	92.8	18.18
Base Money Growth Rate	13.01	18.40	−26.46	207.12	26.35
Taxes (% GDP)	16.37	17.21	0.09	144.12	9.48

Variables for Subsample of Countries Which Have Billionaires at Least Once	Mean	Median	Min	Max	Standard Deviation
Billionaire Wealth (% GDP)	1.95	4.89	0	217.85	13.81
Inherited Billionaire Wealth (% GDP)	0.7	1.77	0	16.44	2.90
Education	69.05	68.09	25.4	92.8	14.25
Base Money Growth Rate	13.01	18.54	−13.55	207.12	27.75
Taxes (% GDP)	16.8	17.66	0.09	48.34	7.27

Countries with Billionaires in 1987	24
Countries with Billionaires in 1992	32
Countries with Billionaires in 1996	41
Countries with Billionaires in 2002	45
Countries with Billionaires in 2007	55
Countries with Billionaires in 2012	59

Variables for the OECD Subsample	Mean	Median	Min	Max	Standard Deviation
Billionaire Wealth (% GDP)	1.60	2.81	0	22.07	3.77
Inherited Billionaire Wealth (% GDP)	0.87	1.72	0	16.44	2.57
Education	78	75.78	39.9	92.8	10.27
Base Money Growth Rate	10.81	16.70	−12.01	192.5	27.14
Taxes (% GDP)	21.03	19.49	1.26	35.09	6.65
Central Bank Target Rate	3.44	4.48	0.07	60.71	6.22
M1 Level	49.89	57.97	0.07	140.74	35.24
Gini Index	0.30	0.31	0.24	0.48	0.05

Notes: Billionaire Wealth as a Percentage of GDP, Inherited Billionaire Wealth, and Education are multiplied by 100 both in these summary statistics and in the regression analysis. Base Money Growth Rate is divided by 10 in the regression analysis but not in these summary statistics.

S2 Robustness Extensions

Table S2.1: Wealth Inequality and Base Money Growth in Countries that Have Had at Least One Billionaire

Panel A	(1)	(2)	(3)
Wealth Inequality	One-Step	One-Step with Robust SE	Two-Step with Robust SE
Base money growth	1.053**	1.053***	1.190***
	(2.55)	(2.94)	(3.97)
First lag of	0.449***	0.449**	0.360
dependent variable	(3.52)	(2.16)	(1.32)
Education	−0.144	−0.144	−0.101
	(−0.50)	(−0.61)	(−0.53)
Tax as a percentage	0.744***	0.744***	0.774***
of GDP	(2.66)	(3.22)	(3.08)
Log GDP per capita	−1.207	−1.207	−1.794
	(−0.22)	(−0.28)	(−0.57)
Constant	15.687	15.687	16.448
	(0.35)	(0.41)	(0.62)
N	159	159	159
Sargan test statistic	0.564	–	–
Arellano-Bond I	–	0.028	0.253
Arellano-Bond II	–	0.510	0.477
Panel B	(1)	(2)	(3)
Inherited Wealth Inequality	One-Step	One-Step with Robust SE	Two-Step with Robust SE
Base money growth	0.714***	0.714***	0.657**
	(4.35)	(2.91)	(2.37)
First lag of	0.312***	0.312**	0.23
dependent variable	(2.70)	(2.07)	(1.02)
Education	−0.020	−0.020	−0.086
	(−0.18)	(−0.17)	(−0.65)
Tax as a percentage	0.181*	0.181*	0.166
of GDP	(1.83)	(1.67)	(1.20)
Log GDP per capita	−1.963	−1.963	−1.455
	(−1.20)	(−0.70)	(−0.62)
Constant	19.074	19.074	18.882
	(1.17)	(0.64)	(0.64)
N	159	159	159
Sargan test statistic	0.624	–	–
Arellano-Bond I	–	0.011	0.195
Arellano-Bond II	–	0.667	0.746

Notes: *Panel A*: dependent variable is billionaire wealth as a percentage of GDP. *Panel B*: dependent variable is *inherited* billionaire wealth as a percentage of GDP. * significant at 10%; ** significant at 5%; *** significant at 1%. Z statistics in parentheses. Education is the UN education index. Base money growth is from the IMF, while M1 is from the OECD. Tax revenue as a percentage of GDP is from the World Bank. GDP per capita is from the Penn World Tables. Arellano-Bond I and II are the Arellano-Bond tests for first-order and second-order autocorrelation.

Table S2.2: Wealth Inequality and Base Money Growth in the OECD

Panel A	(1)	(2)	(3)
Wealth Inequality	One-Step	One-Step with Robust SE	Two-Step with Robust SE
Base money growth	0.996***	0.996***	0.719**
	(5.03)	(3.62)	(2.26)
First lag of	0.445***	0.445	0.608**
dependent variable	(4.19)	(1.33)	(2.52)
Education	0.188	0.188	0.107
	(0.80)	(0.82)	(0.44)
Tax as a percentage	−0.069	−0.069	0.51*
of GDP	(−0.29)	(−0.33)	(1.69)
Log GDP per capita	8.043**	8.043	2.019
	(2.47)	(1.38)	(0.53)
Constant	−97.694***	−97.694	−38.474
	(−2.93)	(−1.59)	(−1.00)
N	99	99	99
Sargan test statistic	0.115	–	–
Arellano-Bond I	–	0.167	0.261
Arellano-Bond II	–	0.523	0.492
Panel B	(1)	(2)	(3)
Inherited Wealth Inequality	One-Step	One-Step with Robust SE	Two-Step with Robust SE
Base money growth	0.730***	0.730***	0.667***
	(5.57)	(5.30)	(4.07)
First lag of	0.350***	0.350	0.438
dependent variable	(3.07)	(0.91)	(1.49)
Education	−0.040	−0.040	0.067
	(−0.24)	(−0.30)	(0.65)
Tax as a percentage	0.156	0.156	0.206
of GDP	(0.97)	(0.87)	(1.56)
Log GDP per capita	7.649***	7.567*	5.485
	(3.08)	(1.68)	(1.24)
Constant	−79.286***	−79.286*	−66.815
	(−2.79)	(−1.85)	(−1.50)
N	99	99	99
Sargan test statistic	0.007	–	–
Arellano-Bond I	–	0.080	0.264
Arellano-Bond II	–	0.387	0.381

Notes: *Panel A*: dependent variable is billionaire wealth as a percentage of GDP. *Panel B*: dependent variable is *inherited* billionaire wealth as a percentage of GDP. * significant at 10%; ** significant at 5%; *** significant at 1%. Z statistics in parentheses. Central bank target rates are from country central bank websites and the ECB for the euro area. See notes at the end of Table S2.1 for details on other regressor variables. Arellano-Bond I and II are the Arellano-Bond tests for first-order and second-order autocorrelation.

Table S2.3: Wealth Inequality and Central Bank Targeted Rates in the OECD

Panel A	(1)	(2)	(3)
Wealth Inequality	One-Step	One-Step with Robust SE	Two-Step with Robust SE
Central bank	0.482**	0.482*	0.392
target rate	(2.07)	(1.65)	(1.33)
First lag of	0.387***	0.387	0.457
dependent variable	(3.56)	(1.14)	(1.12)
Education	0.013	0.013	0.042
	(0.08)	(0.08)	(0.32)
Tax as a percentage	0.288	0.288	0.303
of GDP	(1.41)	(1.57)	(1.32)
Log GDP per capita	7.230**	7.23	5.115
	(2.29)	(1.36)	(1.19)
Constant	−81.081***	−81.081*	−61.916
	(−2.61)	(−1.66)	(−1.62)
N	119	119	119
Sargan test statistic	0.237	–	–
Arellano-Bond I	–	0.069	0.348
Arellano-Bond II	–	0.333	0.378
Panel B	(1)	(2)	(3)
Inherited Wealth Inequality	One-Step	One-Step with Robust SE	Two-Step with Robust SE
Central bank	0.384**	0.384*	0.369
target rate	(2.24)	(1.84)	(1.33)
First lag of	0.331***	0.331	0.492*
dependent variable	(2.65)	(0.89)	(1.82)
Education	0.058	0.058	0.112
	(0.42)	(0.65)	(1.15)
Tax as a percentage	0.204	0.204	0.0892
of GDP	(1.52)	(1.56)	(0.66)
Log GDP per capita	8.348***	8.348*	5.728
	(3.09)	(1.79)	(1.05)
Constant	−95.912***	−95.912**	−91.532**)
	(−3.21)	(−2.06)	(−2.08)
N	116	116	116
Sargan test statistic	0.296	–	–
Arellano-Bond I	–	0.047	0.338
Arellano-Bond II	–	0.204	0.483

Notes: *Panel A*: dependent variable is billionaire wealth as a percentage of GDP. *Panel B*: dependent variable is *inherited* billionaire wealth as a percentage of GDP. * significant at 10%; ** significant at 5%; *** significant at 1%. Z statistics in parentheses. Central bank target rates are from country central bank websites and the ECB for the euro area. See notes at the end of Table S2.1 for details on other regressor variables. Arellano-Bond I and II are the Arellano-Bond tests for first-order and second-order autocorrelation.

Table S2.4: Robustness Extensions for OECD: Income Inequality (Gini) and Level of M1 Money Supply

	(1)	(2)	(3)
	Fixed Effects OLS	One-Step with Robust SE	Two-Step with Robust SE
Dependent variable:	Gini	B/GDP	B/GDP
Base money growth	0.122		
	(1.36)		
M1 level		−0.656	−3.973
		(−0.16)	(−1.01)
First lag of	–	0.424	0.400
dependent variable		(1.42)	(1.21)
Education	−0.046	−0.308*	−0.227**
	(−0.17)	(−1.75)	(−2.30)
Tax as a percentage	0.293	0.108	0.169
of GDP	(1.50)	(0.81)	(1.60)
Log GDP per capita	−1.508	4.264	3.413
	(−0.45)	(1.13)	(1.10)
Constant	44.621	−18.236	−13.361
	(1.22)	(−0.59)	(−0.53)
N	52	154	154
Sargan test statistic	–	–	–
Arellano-Bond I	–	0.078	0.317
Arellano-Bond II	–	0.158	0.296

Notes: * significant at 10%; ** significant at 5%; *** significant at 1%. Z statistics in parentheses. Dependent variable in column (1) is the Gini index, which comes from the OECD, while dependent variable in columns (2) and (3) is billionaire wealth as a percentage of GDP. M1 is from the OECD. Education is the UN education index. Base money growth is from the IMF. Tax revenue as a percentage of GDP is from the World Bank. GDP per capita is from the Penn World Tables. Arellano-Bond I and II are the Arellano-Bond tests for first-order and second-order autocorrelation.

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